

Homework 1: Green Machines

Year 9 Science · about 40 minutes

Section A: A Section B: M / E

Section A

1 Write the meaning of each word. [A]

(a) germination

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(b) transpiration

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(c) zygote

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2 Complete the sentences about flower parts. [A]

The _____ protects the flower while it is a bud. The _____ makes sweet nectar. The _____ is sticky so that pollen grains stay on it.

3 Name three things a seedling competes with its parent plant for. [A]

1. _____ 2. _____ 3. _____

Quick review

4 On a hot, dry afternoon a gardener sees the leaves of her tomato plants drooping. [A]

(a) Which process is the plant losing water by? _____

(b) Through which openings in the leaf is the water lost? _____

(c) Which tubes should be bringing water up from the roots? _____

Section B

5 Explain how a bee pollinates a flower. Use the names of the flower parts. [M]

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6 Explain why cross-pollination is better for a species than self-pollination. [E]

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7 Gardeners often soak bean seeds in water overnight before planting them. Explain why. [M]

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8 A carrot left in fresh water for a day becomes firm. A carrot left in very salty water becomes soft and bendy. Explain both results using osmosis. [E]

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9 A potted plant is weighed each day. All the mass it loses is water. [M]

Day	1	2	3	4
Mass (g)	500	488	476	464

(a) How much water does the plant lose each day?

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(b) Explain where the water goes.

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10 Sam explains how water gets to the leaves of a plant. [E]

Sam's answer

Water enters the root hairs by osmosis, from a low concentration of water to a high concentration of water. It travels up the phloem to the leaves. The leaves make glucose and oxygen from carbon dioxide and water. At night the plant stops respiring.

Sam's answer has 3 errors. Find each one, fix it, and say why it loses marks.

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Also set: Green Machines booklet p.24, fill in the glossary for any ten words.